1.6 Finite temperature and Matsubara Green functions

Consider translation invariance in time domain (equilibrium systems, time-independent H, ...), causal GF:

$$G(\vec{r},t,t') = G(\vec{r},t-t') = G(\vec{r},t) = -i\langle T_+ c_{\vec{r}}(t)\, c_{\vec{0}}^\dagger(0)\rangle$$

In the time domain particular care should be taken because of ① time-ordered product, ② finite-T cases

In fact, at finite T:

$$G(\vec{r},t) = -\frac{i}{Z}\,\mathrm{Tr}\left[e^{-\beta H}\, T_+\, \psi(\vec{r},t)\,\psi^\dagger(0,0)\right] =$$

$$= -\frac{i}{Z}\,\mathrm{Tr}\left[T_+\, \underbrace{e^{-\beta H}\, e^{iHt}}\, \psi(\vec{r})\, e^{-iHt}\,\psi(0)\right]$$

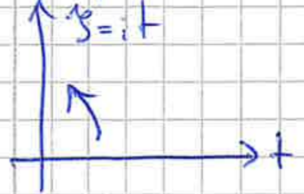
⇒ evolution with a "complex" time argument ($\tilde{t} = -\beta + it$)

This may lead to many undesired complications, which can be avoided using the Matsubara formalism (*)

(*) T. Matsubara (1921–2014)

We first perform a Wick rotation (*) of the time axis

(*) G.C. Wick (1909–1992)



$\Rightarrow \tau = it$ (so that $\tilde{t} = -\beta + \tau$)

"imaginary time"

$$\Rightarrow G(\vec{r},\tau) = -\langle T_\tau\, \psi(\vec{r},\tau)\,\psi^\dagger(0,0)\rangle = -\frac{1}{Z}\,\mathrm{Tr}\left[e^{-\beta H}\, T_\tau\, \psi(\vec{r},\tau)\,\psi^\dagger(0,0)\right]$$

This poses, however, precise restrictions to the definition of τ:

$$\begin{cases} \tau>0: \ \mathrm{Tr}\left[e^{-\beta H} e^{H\tau}\psi(\vec r)\dots\right] = \sum_n e^{-(\beta-\tau)E_n}\dots \Rightarrow 0<\tau<\beta \\ \tau<0: \ \mathrm{Tr}\left[e^{-\beta H}\psi^\dagger(0)\dots\psi(\vec r)e^{-H\tau}\right] = \sum_n e^{-(\beta+\tau)E_n} \Rightarrow -\beta<\tau<0 \end{cases}$$

$$\Rightarrow \quad \boxed{-\beta<\tau<\beta}$$

Moreover, we can exploit the cyclic properties of the trace and the sign-change for T_τ for fermions:

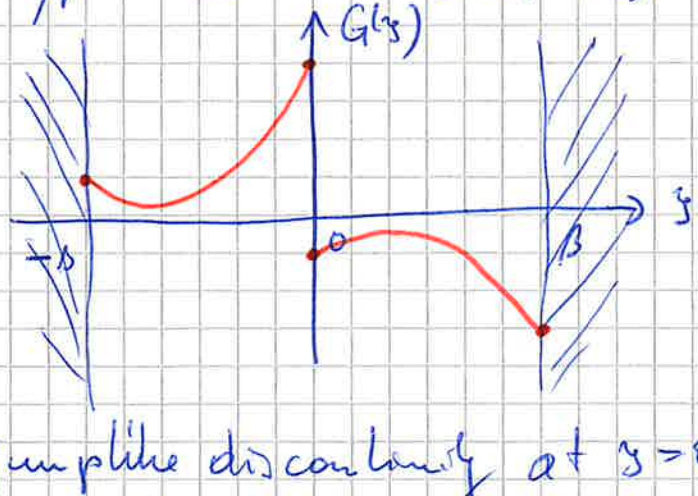
$$\tau<0: \quad G(\vec r,\tau) = +\frac{1}{Z}\mathrm{Tr}\left[e^{-\beta H}\psi^\dagger(0,0)\psi(\vec r,\tau)\right] =$$

$$= \frac{1}{Z}\mathrm{Tr}\left[e^{-\beta H}\psi^\dagger(0,0)\, e^{H\tau}\psi(\vec r,0)e^{-H\tau}\right] =$$

$$= \frac{1}{Z}\mathrm{Tr}\left[e^{H\tau}\psi(\vec r,0)e^{-(\tau+\beta)H}\psi^\dagger(0,0)\right] =$$

$$= \frac{1}{Z}\mathrm{Tr}\left[e^{-\beta H}e^{H(\tau+\beta)}\psi(\vec r,0)e^{-H(\tau+\beta)}\psi^\dagger(0,0)\right] = -G(\vec r,\tau+\beta)$$

Hence, the typical behavior of $G(\vec r,\tau)$ is the following:

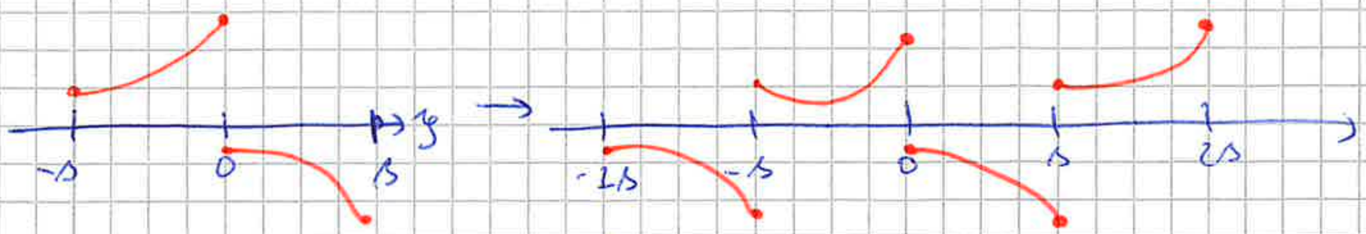


with a jumplike discontinuity at $\tau=0$

local dens. of system

$$\text{e.g.: } G(\vec r=0,\tau=0) = \begin{cases} \tau=0^+: & -\langle\psi(\vec r,0)\psi^\dagger(0,0)\rangle \xRightarrow[\vec r=0]{} -1+n(\vec r=0) \\ \tau=0^-: & \langle\psi^\dagger(0,0)\psi(\vec r,0)\rangle \Rightarrow n(\vec r=0) \end{cases}$$

As $G(\tau)$ is defined on a finite interval, its Fourier transform is obtained by periodization on the whole τ-axis:



Hence:
$$\begin{cases} G(\vec{k}, i\omega_n) = \frac{1}{2}\int_{-\beta}^{\beta} d\tau\, G(\vec{k},\tau)\, e^{i\omega_n \tau} \\ G(\vec{k},\tau) = \frac{1}{\beta}\sum_n G(\vec{k}, i\omega_n)\, e^{-i\omega_n \tau} \end{cases}$$

where the allowed frequencies would be $\omega_n = \frac{2\pi}{2\beta} n$, $n \in \mathbb{Z}$

Because of the cyclic property of the trace $[G(\vec{k},\tau) = -G(\vec{k},\tau+\beta), \tau<0]$, a further reduction can be found:

$$G(\vec{k}, i\omega_n) = \frac{1}{2}\int_{-\beta}^{0} d\tau\, \underbrace{G(\vec{k},\tau)}_{= -G(\vec{k},\tau+\beta)}\, e^{i\omega_n \tau} + \frac{1}{2}\int_{0}^{\beta} d\tau\, G(\vec{k},\tau)\, e^{i\omega_n \tau} =$$

$$= -\frac{1}{2}\int_{0}^{\beta} d\tau'\, G(\vec{k},\tau')\, e^{i\omega_n(\tau'-\beta)} + \frac{1}{2}\int_{0}^{\beta} d\tau\, G(\vec{k},\tau)\, e^{i\omega_n \tau} =$$

$$= \frac{1}{2}\int_{0}^{\beta} d\tau\, G(\vec{k},\tau)\, e^{i\omega_n \tau}\left[1 - e^{-i\omega_n \beta}\right]$$

0 if n is even

$e^{-i\frac{\pi}{\beta} n\beta} = (-1)^n$

$1-(-1) = \underline{\underline{2}}$

$$\Rightarrow \begin{cases} G(\vec{k}, i\omega_n) = \int_{0}^{\beta} d\tau\, G(\vec{k},\tau)\, e^{i\omega_n \tau} \\ G(\vec{k},\tau) = \frac{1}{\beta}\sum_n G(\vec{k}, i\omega_n)\, e^{-i\omega_n \tau} \end{cases}$$

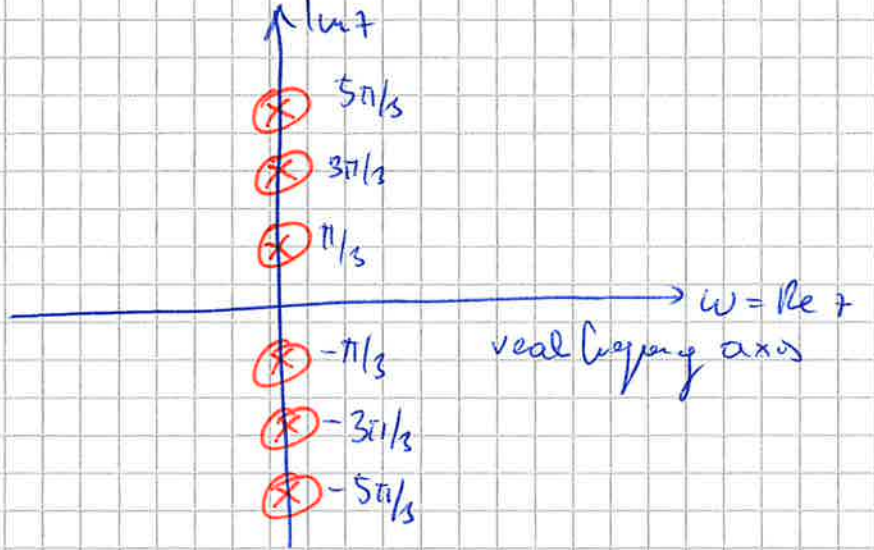
with $\omega_n = (2n+1)\frac{\pi}{\beta}$, $n \in \mathbb{Z}$

fermionic Matsubara frequencies

(for bosons: $\omega_n = 2n\pi/\beta$)

Visualization and interpretation

$$z = \underbrace{\mathrm{Re}\,z}_{\omega} + i\,\mathrm{Im}\,z$$



Interpretation: $n_F(\varepsilon) = \dfrac{1}{e^{\beta\varepsilon}+1} \xrightarrow[\text{plane}]{\text{complex}} \dfrac{1}{e^{\beta z}+1}$

poles of $f(z)$ $\Rightarrow$ $(e^{\beta z}+1) \overset{!}{=} 0 \Rightarrow z = i(2n+1)\frac{\pi}{\beta} =$
$= i\omega_n$

$\Rightarrow$ the poles of the Fermi-Dirac distribution on the complex plane are the fermionic Matsubara frequencies

Non-interacting Matsubara Green function

Substitution from real times

$$H = \sum_{\vec{k}\sigma} \xi_{\vec{k}}\, c^\dagger_{\vec{k}\sigma} c_{\vec{k}\sigma}$$

$\tau > 0$: $G^{(0)}(\vec{k},\tau) = -\langle c_{\vec{k}}(\tau) c^\dagger_{\vec{k}}(0)\rangle = -e^{-\xi_{\vec{k}}\tau}\left(1 - n_F(\xi_k)\right)$

~~Fourier transform~~

Special case: only one atomic energy level ε

$$\Rightarrow G^{(0)}(\tau) = -e^{-\varepsilon\tau}\left(1 - n_F(\varepsilon)\right)$$

Fourier transform:

$$\boxed{G^{(0)}(\vec{k}, i\omega_n)} = \int_0^\beta d\tau\, e^{-\xi_{\vec{k}}\tau}\left[n_F(\xi_{\vec{k}})-1\right] e^{i\omega_n\tau} =$$

$$= \frac{[f(\xi_{\vec{k}})-1]}{i\omega_n-\xi_{\vec{k}}}\, e^{(i\omega_n-\xi_{\vec{k}})\tau}\Big|_{\tau=0}^{\tau=\beta} =$$

$$= \frac{1}{i\omega_n-\xi_{\vec{k}}}\Big[\underbrace{\cancel{n_F(\xi_{\vec{k}})}}_{=\frac{1}{e^{\beta\xi_k}+1}}\cancel{-1}\Big]\left[\cancel{-e^{-\beta\xi_k}}-1\right] = \boxed{\frac{1}{i\omega_n-\xi_{\vec{k}}}}$$

Lehmann representation:

$$G(\vec{k},\tau) = -\frac{1}{Z}\sum_N \langle N| e^{-\beta E_N} c_k(\tau) c^+_{\vec{k}}(0)|N\rangle =$$

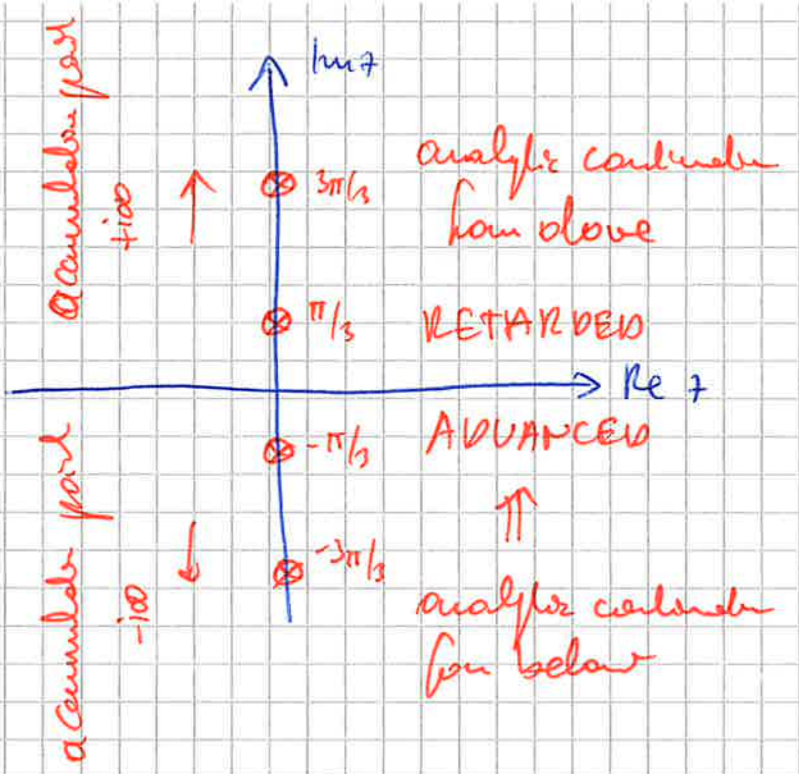
$$= -\frac{1}{Z}\sum_{N,M} \langle N| e^{-\beta E_N} e^{\tau E_N} c_{\vec{k}} e^{-\tau E_M}|M\rangle\langle M| c^+_{\vec{k}}|N\rangle =$$

$$= -\frac{1}{Z}\sum_{N,M} e^{-\beta E_N} e^{\tau(E_N-E_M)} |\langle N|c_{\vec{k}}|M\rangle|^2$$

Fourier transform: $G(\vec{k}, i\omega_n) = \frac{1}{Z}\sum_{N,M}\frac{|\langle N|c_k|M\rangle|^2 (e^{-\beta E_M}+e^{-\beta E_N})}{i\omega_n-(E_M-E_N)}$

analytic continuation on real frequencies ($i\omega_n \to \omega$)

$G(\vec{k}, i\omega_n)$ is known on a discrete set of frequencies.
However ~~to~~ to guarantee the uniqueness of its analytic continuation at least one accumulation point is needed



in particular $G(\vec{k}, z = \omega + i0)$ corresponds to the retarded Green function

$$G_R(\vec{k},\omega) = \frac{1}{Z} \sum_{n,m} \frac{|\langle n|c_k|m\rangle|^2 \left(e^{-\beta E_m} + e^{-\beta E_n}\right)}{\omega + i0^+ - (E_m - E_n)}$$